T3 A_-4: general EFT1 RGEs

Matchete/RGBeta symbolic output in mathematical notation

A_-4

Additional quartics: 0, additional bilinears: 0, extra Yukawa: none.
General matching and Wilson coefficients are not computed in this symbolic report.

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{55{g_1}^3}{6}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{5{g_2}^3}{2}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} = &3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij}\\ &+3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij}\\ &+\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij}\\ &+\frac{3}{2}\Big(Y_e\ (Y_e)^\dagger\ Y_e\Big)_{ij}\\ &-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}\\ &-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

m_1^2

$$\begin{aligned}16\pi^2 \frac{dm_1^2}{d\ln\mu} = &-24{g_1}^2 m_1^2\\ &+6\lambda_{12}^{(1)} m_2^2\\ &+4\lambda_{S_1}^{(1)} m_1^2\end{aligned}$$

m_2^2

$$\begin{aligned}16\pi^2 \frac{dm_2^2}{d\ln\mu} = &-12{g_2}^2 m_2^2\\ &-6{g_1}^2 m_2^2\\ &+2\lambda_{12}^{(1)} m_1^2\\ &+m_2^2(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})\end{aligned}$$

λ_1

$$\begin{aligned}16\pi^2 \frac{d\lambda_1}{d\ln\mu} = &\lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))\\ &-12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)\\ &-4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\ &-12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)\\ &+4\lambda_5\,(\lambda_5)^*\\ &+\frac{9{g_2}^4}{4}\\ &+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)\\ &-9{g_2}^2\lambda_1\\ &+\frac{3{g_1}^4}{4}\\ &+12{\lambda_1}^2\\ &+2{\lambda_{HS_1}^{(1)}}^2\end{aligned}$$

$$
\begin{aligned}
&+ 6{\lambda_{HS_2}^{(1)}}^2 \\
&+ 3{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = &192{g_1}^4 \\
&- 48{g_1}^2\lambda_{S_1}^{(1)} \\
&+ 6{\lambda_{12}^{(1)}}^2 \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 10{\lambda_{S_1}^{(1)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = &22{g_2}^4 \\
&+ {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_2}^{(1)}\right) \\
&- 24{g_2}^2\lambda_{S_2}^{(1)} \\
&+ 12{g_1}^4 \\
&+ 2{\lambda_{12}^{(1)}}^2 \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 14{\lambda_{S_2}^{(1)}}^2 \\
&+ 8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 16{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = &\lambda_{HS_1}^{(1)}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)) \\
&+ 12\lambda_5\,(\lambda_5)^* \\
&- \frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 12{g_1}^4 \\
&- \frac{51{g_1}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = &\lambda_{HS_2}^{(1)}(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_2}^{(A)}) \\ &+ 4\lambda_5\,(\lambda_5)^* \\ &+ 6{g_2}^4 \\ &- \frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2} \\ &+ 3{g_1}^4 \\ &- \frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2} \\ &+ 2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\ &+ 6\lambda_1\lambda_{HS_2}^{(1)} \\ &+ 4{\lambda_{HS_2}^{(1)}}^2 \\ &+ 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\ &+ 6{\lambda_{HS_2}^{(A)}}^2\end{aligned}$$

$\lambda_{12}^{(1)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = &4\lambda_5\,(\lambda_5)^* \\ &- 12{g_2}^2\lambda_{12}^{(1)} \\ &+ 48{g_1}^4 \\ &- 30{g_1}^2\lambda_{12}^{(1)} \\ &+ 4{\lambda_{12}^{(1)}}^2 \\ &+ 4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\ &+ 8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\ &+ 4\lambda_{12}^{(1)}\lambda_{S_2}^{(A)} \\ &+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}\end{aligned}$$

λ_5

$$\begin{aligned}16\pi^2 \frac{d\lambda_5}{d\ln\mu} = &\lambda_5(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{HS_2}^{(A)}) \\ &- \frac{21{g_2}^2\lambda_5}{2} \\ &- \frac{33{g_1}^2\lambda_5}{2} \\ &+ 2\lambda_{12}^{(1)}\lambda_5 \\ &+ 2\lambda_1\lambda_5 \\ &+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\ &+ 4\lambda_{HS_2}^{(1)}\lambda_5\end{aligned}$$

$\lambda_{HS_2}^{(A)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = \lambda_{HS_2}^{(A)}(&6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_2}^{(A)}) \\ &+ \frac{8}{3}\lambda_5\,(\lambda_5)^* \\ &+ {g_1}^2\left(-4{g_2}^2 - \frac{15\lambda_{HS_2}^{(A)}}{2}\right) \\ &- \frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2} \\ &+ 2\lambda_1\lambda_{HS_2}^{(A)} \\ &+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\ &+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}\end{aligned}$$

$\lambda_{S_2}^{(A)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{S_2}^{(A)}}{d\ln\mu} = &-2{g_2}^4 \\ &+ {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_2}^{(A)}\right) \\ &- 24{g_2}^2\lambda_{S_2}^{(A)} \\ &+ 3{\lambda_{HS_2}^{(A)}}^2 \\ &+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\ &+ 20{\lambda_{S_2}^{(A)}}^2\end{aligned}$$